Module 2: Electrostatic bridge from radiation defects to field

Purpose: convert mobile carriers, ionized dopants, and charged radiation-induced defects into a self-consistent electrostatic potential and electric field in silicon.

Space-charge density

$$\boxed{\rho = q(p - n + N_D^+ - N_A^-) + \rho_{\mathrm{def}}}$$

- p: mobile hole concentration.
- n: mobile electron concentration.
- N_D^+: ionized donor concentration.
- N_A^-: ionized acceptor concentration.
- ρ_{def}: charge from radiation-induced defect populations.

Electrostatic output

$$\boxed{\rho(x, y, t) \longrightarrow \phi(x, y, t) \longrightarrow \boldsymbol{E}(x, y, t)}$$

$$\boxed{\boldsymbol{E} = -\nabla\phi}$$

The module does not evolve defects by itself. It receives defect populations from Module 3, converts them into charge density, then solves the field problem.

Core message: radiation damage changes material charge; charge bends potential; potential gradients create the electric field that controls drift in Module 5.

Module 2: Electrostatic limit and Poisson equation

Module 2 uses electroquasistatics: the field is charge-dominated, not induction-dominated.

Maxwell starting point

$$\begin{aligned}\nabla \cdot \boldsymbol{D} &= \rho,\\ \boldsymbol{D} &= \epsilon \boldsymbol{E},\\ \boldsymbol{E} &= -\nabla \phi.\end{aligned}$$

Substitution gives the electrostatic field equation:

$$\boxed{\nabla \cdot (\epsilon \nabla \phi) = -\rho}$$

For uniform silicon permittivity,

$$\boxed{\epsilon_{\mathrm{Si}} \nabla^2 \phi = -\rho.}$$

Why induction is neglected

$$\boxed{\frac{E_{\mathrm{ind}}}{E_{\mathrm{es}}} \sim \left(\frac{L}{c_{\mathrm{Si}} \tau}\right)^2}$$

- L: device length scale.
- τ: time scale for charge redistribution.
- $c_{\mathrm{Si}} = 1/\sqrt{\mu \epsilon}$: EM wave speed in silicon.

If $\tau \gg L/c_{\mathrm{Si}}$, the field adjusts much faster than the material charge evolves; Poisson's equation is the correct reduced model.

Field solver equation: $\dfrac{\partial^2 \phi}{\partial x^2} + \dfrac{\partial^2 \phi}{\partial y^2} = -\rho/\epsilon_{\mathrm{Si}}.$

Module 2: Defect species and effective charge state

A defect species is a radiation-induced material-defect population, not a mobile carrier or dopant.

Defect-charge map

$$\boxed{\rho_{\mathrm{def}} = q\sum_{i=1}^{M} z_i C_i}$$

Examples of C_i:

Field	Meaning	Charge factor
C_V	vacancy concentration	z_V
C_I	self-interstitial concentration	z_I
C_{V_2}	divacancy concentration	z_{V_2}
C_{VO}	vacancy-oxygen complex	z_{VO}

These are distinct from p, n, N_D^+, N_A^-, which already appear separately in ρ.

Why a vacancy is not automatically $+q$ A vacancy is a missing lattice atom, not a missing valence-band electron. Removing a silicon atom changes a covalent bonding network: dangling bonds, lattice relaxation, and localized defect states may appear.

$$\boxed{V^+, \quad V^0, \quad V^-, \quad V^{2-}, \ldots}$$

The dimensionless factor z_i represents the net electronic charge state after defect-state occupancy, not simply the fact that an atom is missing.

$z_i > 0$: donor-like contribution; $z_i < 0$: acceptor-like contribution; $z_i = 0$: neutral in the reduced model.

Module 2: FEM weak form and matrix equation

The continuous Poisson equation becomes a finite-dimensional system for nodal potentials.

Weak form

$$\nabla \cdot (\epsilon_{\mathrm{Si}} \nabla \phi) = -\rho$$

Multiply by test function w, integrate by parts:

$$\boxed{\int_\Omega \epsilon_{\mathrm{Si}} \nabla w \cdot \nabla \phi \, \mathrm{d}\Omega = \int_\Omega w \rho \, \mathrm{d}\Omega + \int_{\Gamma_N} w \bar{d} \, \mathrm{d}\Gamma}$$

where $\bar{d} = \boldsymbol{n} \cdot \epsilon_{\mathrm{Si}} \nabla \phi$ is prescribed normal displacement-flux data.

Linear triangular element

$$\phi_h(x, y) = \sum_{a=1}^{3} N_a(x, y) \Phi_a, \qquad w_h = N_b.$$

$$\boxed{K^e_{ab} = \int_{\Omega_e} \epsilon_{\mathrm{Si}} \nabla N_a \cdot \nabla N_b \, \mathrm{d}\Omega}$$

$$\boxed{b^e_a = \int_{\Omega_e} N_a \rho \, \mathrm{d}\Omega + \int_{\Gamma^e_N} N_a \bar{d} \, \mathrm{d}\Gamma}$$

$$\boxed{\mathbf{K} \boldsymbol{\Phi} = \boldsymbol{b}}$$

Dirichlet contacts prescribe ϕ; Neumann/symmetry boundaries prescribe normal field. Postprocess $\boldsymbol{E} = -\nabla \phi$ elementwise or at nodes.

Module 3: Defect evolution equation and source terms

Purpose: evolve radiation-induced defect populations from local generation, diffusion, reaction, clustering, and annealing.

Single-species reduced model

$$\boxed{\frac{\partial C}{\partial t} = \nabla \cdot (D \nabla C) + G - k_a C - R(C)}$$

- $C(x, y, t)$: defect concentration.
- D: effective diffusivity.
- G: radiation-induced generation source.
- $k_a C$: first-order annealing/removal.
- $R(C)$: nonlinear reaction or clustering term.

Multi-species form

$$\boxed{\frac{\partial C_i}{\partial t} = \nabla \cdot (D_i \nabla C_i) + G_i + \sum_r \nu_{ir} \mathcal{R}_r(\{C_j\}) - k_i C_i}$$

A reaction r may represent recombination, complex formation, dissociation, or trap creation.

$$V + I \to \emptyset, \qquad V + O \to VO, \qquad V + V \to V_2.$$

Module handoff: Module 3 outputs $C_i(x, y, t)$; Module 2 maps them to $\rho_{\text{def}} = q \sum_i z_i C_i$.

Module 3: Effective transport and kinetic coefficients

The defect PDE is simple in form, but its coefficients carry the material physics.

Thermally activated motion

$$\boxed{D_i(T) = D_{i0} \exp\left(-\frac{E_{m,i}}{k_B T}\right)}$$

$$\boxed{k_i(T) = \nu_i \exp\left(-\frac{E_{a,i}}{k_B T}\right)}$$

$E_{m,i}$ is a migration barrier; $E_{a,i}$ is an annealing or reaction activation barrier.

Possible field dependence If charged-defect drift is included,

$$\boxed{\boldsymbol{J}_i = -D_i \nabla C_i + \mu_i z_i q C_i \boldsymbol{E}}$$

$$\boxed{\frac{\partial C_i}{\partial t} = -\nabla \cdot \boldsymbol{J}_i + G_i + \text{reactions}}$$

Einstein relation, when applicable:

$$\boxed{\mu_i = \frac{D_i}{k_B T}}$$

The reduced baseline may omit this drift term; the coupled model can feed $\boldsymbol{E}$ from Module 2 back into Module 3.

Interpretation: Module 3 is where radiation damage becomes an evolving material state, not merely a static fluence-to-defect conversion.

Module 3: FEM weak form for diffusion–reaction defects

The FEM form converts a defect transport PDE into mass, diffusion, reaction, and source matrices.

Weak form

$$\frac{\partial C}{\partial t} = \nabla\cdot(D\nabla C) + G - k_a C$$

Multiply by w, integrate by parts:

$$\boxed{\int_\Omega w\frac{\partial C}{\partial t}\,\mathrm{d}\Omega + \int_\Omega D\nabla w\cdot\nabla C\,\mathrm{d}\Omega + \int_\Omega w k_a C\,\mathrm{d}\Omega = \int_\Omega wG\,\mathrm{d}\Omega}$$

Natural zero-flux boundary:

$$\boxed{\boldsymbol{n}\cdot D\nabla C = 0}$$

Element matrices

$$C_h = \sum_a N_a C_a$$

$$\begin{aligned} M_{ab}^e &= \int_{\Omega_e} N_a N_b\,\mathrm{d}\Omega, \\ K_{ab}^e &= \int_{\Omega_e} D\nabla N_a\cdot\nabla N_b\,\mathrm{d}\Omega, \\ R_{ab}^e &= \int_{\Omega_e} k_a N_a N_b\,\mathrm{d}\Omega, \\ g_a^e &= \int_{\Omega_e} N_a G\,\mathrm{d}\Omega. \end{aligned}$$

$$\boxed{\mathbf{M}\dot{\boldsymbol{C}} + (\mathbf{K}+\mathbf{R})\boldsymbol{C} = \boldsymbol{g}}$$

Module 3: Time stepping, verification, and coupling

Backward Euler gives a stable update for diffusion and annealing over stiff time scales.

Implicit defect update

$$\boxed{[\mathbf{M} + \Delta t(\mathbf{K} + \mathbf{R})]\,\boldsymbol{C}^{n+1} = \mathbf{M}\boldsymbol{C}^{n} + \Delta t\,\boldsymbol{g}^{n+1}}$$

For nonlinear reactions,

$$\boxed{\mathbf{F}(\boldsymbol{C}^{n+1}) = \mathbf{0}}$$

may be solved by fixed-point iteration, Newton iteration, or operator splitting.

Verification logic

- **Pure annealing:** with $D = 0, G = 0$, recover $C(t) = C_0 e^{-k_a t}$.
- **Uniform preservation:** with zero flux and no source, uniform C remains uniform.
- **Diffusion inventory:** with zero flux and no reactions, $\int_\Omega C \,\mathrm{d}\Omega$ is conserved.
- **Coupling sanity:** increasing C_i should increase ρ_{def} in Module 2 according to qz_iC_i.

Module 3 supplies the evolving damage fields that drive electrostatics, thermal coefficients, and recombination/trapping in later modules.

Module 4: Thermal response and ballistic–diffusive motivation

Purpose: convert deposited and thermalized radiation energy into an evolving temperature field, while retaining quasi-ballistic corrections when local Fourier transport is insufficient.

Continuum energy balance

$$\boxed{C_T \frac{\partial T}{\partial t} + \nabla \cdot \boldsymbol{q} = Q_T}$$

Fourier closure:

$$\boxed{\boldsymbol{q} = -k \nabla T}$$

so

$$\boxed{C_T \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_T.}$$

When Fourier is not enough

$$\boxed{\mathrm{Kn} = \frac{\lambda}{L}}$$

- λ: heat-carrier mean free path.
- L: thermal length scale or device dimension.
- $\mathrm{Kn} \ll 1$: diffusive limit.
- $\mathrm{Kn} \sim 0.1 - 1$: ballistic memory matters.

Radiation energy can initially reside in nonlocal carriers before relaxing into the thermal bath.

Module 4 outputs $T(x, y, t)$, which controls defect kinetics, mobilities, recombination rates, and coupled thermal feedback.

Module 4: Reduced ballistic–diffusive source structure

The reduced model keeps a heat-equation solve but modifies the source with ballistic energy exchange.

Energy and flux split

$$\boxed{u = u_b + u_m}, \qquad \boxed{\boldsymbol{q} = \boldsymbol{q}_b + \boldsymbol{q}_m}$$

$$\boxed{u_m = C_T T_m}, \qquad \boxed{\boldsymbol{q}_m = -k\nabla T_m}$$

The subscript b denotes ballistic/nonlocal carrier energy; m denotes medium thermal energy.

Effective medium-temperature PDE

$$\boxed{C_T \frac{\partial T_m}{\partial t} = \nabla \cdot (k\nabla T_m) + Q_{\text{th}} - \nabla \cdot \boldsymbol{q}_b - \frac{\partial u_b}{\partial t}}$$

The ballistic correction can be treated as an effective source:

$$\boxed{Q_{\text{eff}} = Q_{\text{th}} - \nabla \cdot \boldsymbol{q}_b - \partial_t u_b}$$

$$\boxed{C_T \partial_t T_m = \nabla \cdot (k\nabla T_m) + Q_{\text{eff}}}$$

The FEM implementation can solve the same parabolic structure as Fourier heat conduction, while the source term stores the ballistic physics.

Module 4: FEM weak form and thermal matrices

Temperature is approximated on the shared 2D mesh using the same finite-element machinery as the other field modules.

Weak form

$$C_T \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q_{\text{eff}}$$

$$\boxed{\int_\Omega w C_T \frac{\partial T}{\partial t}\, \mathrm{d}\Omega + \int_\Omega k \nabla w \cdot \nabla T\, \mathrm{d}\Omega = \int_\Omega w Q_{\text{eff}}\, \mathrm{d}\Omega + \int_{\Gamma_N} w \bar{q}\, \mathrm{d}\Gamma}$$

Natural insulated boundary:

$$\boxed{\boldsymbol{n} \cdot k \nabla T = 0.}$$

Element arrays

$$T_h = \sum_a N_a T_a$$

$$M_{ab}^e = \int_{\Omega_e} C_T N_a N_b\, \mathrm{d}\Omega,$$

$$K_{ab}^e = \int_{\Omega_e} k \nabla N_a \cdot \nabla N_b\, \mathrm{d}\Omega,$$

$$f_a^e = \int_{\Omega_e} N_a Q_{\text{eff}}\, \mathrm{d}\Omega.$$

$$\boxed{\mathbf{M}_T \dot{\boldsymbol{T}} + \mathbf{K}_T \boldsymbol{T} = \boldsymbol{f}_T}$$

Module 4: Thermal time stepping and checks

The thermal solve advances temperature with sources from radiation thermalization and ballistic energy exchange.

Backward-Euler update

$$\boxed{(\mathbf{M}_T + \Delta t \mathbf{K}_T)\,\boldsymbol{T}^{n+1} = \mathbf{M}_T \boldsymbol{T}^n + \Delta t\, \boldsymbol{f}_T^{n+1}}$$

Source construction

$$\boxed{Q_{\mathrm{th}} = \eta_{\mathrm{th}} \dot{q}_{\mathrm{rad}}^{2D} + Q_{\mathrm{device}}}$$

$$\boxed{Q_{\mathrm{eff}} = Q_{\mathrm{th}} + Q_{\mathrm{ballistic}}}$$

Verification logic

- **Uniform equilibrium:** no source and insulated boundaries preserve uniform T.
- **Fourier Gaussian diffusion:** in the ballistic-free limit, spreading follows diffusion scaling.
- **Uniform source:** spatially uniform Q causes uniform temperature rise.
- **Energy check:** integrated temperature rise matches input energy when heat loss is absent.

Coupling: T modifies defect diffusivity/annealing in Module 3 and carrier mobility/recombination in Module 5.

Module 5: Carrier drift–diffusion with radiation defects

Purpose: evolve mobile electrons and holes under diffusion, electric-field drift, generation, and defect-assisted recombination/trapping.

Continuity equations

$$\boxed{\frac{\partial n}{\partial t}} \boxed{= \frac{1}{q}\nabla \cdot \boldsymbol{J}_n + G_n - R_n},$$

$$\boxed{\frac{\partial p}{\partial t}} \boxed{= -\frac{1}{q}\nabla \cdot \boldsymbol{J}_p + G_p - R_p.}$$

The sign difference follows from electron negative charge and hole positive charge.

Current densities

$$\boxed{\boldsymbol{J}_n} \boxed{= q\mu_n n\boldsymbol{E} + qD_n\nabla n},$$

$$\boxed{\boldsymbol{J}_p} \boxed{= q\mu_p p\boldsymbol{E} - qD_p\nabla p.}$$

$$\boxed{\boldsymbol{E} = -\nabla\phi}$$

Module 2 supplies ϕ and $\boldsymbol{E}$; Module 3 supplies defect densities that influence recombination/trapping.

Carrier transport is where radiation-induced field distortion and defect-assisted recombination become electrical degradation observables.

Module 5: Recombination, generation, and defect coupling

Defects affect mobile carriers both electrostatically and kinetically.

Reduced recombination/trapping model

$$\boxed{R_n \approx \frac{n - n_{\mathrm{eq}}}{\tau_n(C_i, T)}}, \qquad \boxed{R_p \approx \frac{p - p_{\mathrm{eq}}}{\tau_p(C_i, T)}}$$

Defect concentrations can shorten lifetimes:

$$\boxed{\frac{1}{\tau_n} = \frac{1}{\tau_{n0}} + \sum_i \alpha_{n,i} C_i}$$

$$\boxed{\frac{1}{\tau_p} = \frac{1}{\tau_{p0}} + \sum_i \alpha_{p,i} C_i.}$$

Two damage pathways

- **Field pathway:** $C_i \to \rho_{\mathrm{def}} \to \phi, \boldsymbol{E} \to$ drift.
- **Lifetime pathway:** $C_i \to \tau_n, \tau_p \to R_n, R_p \to$ carrier loss.
- **Thermal pathway:** $T \to \mu_n, \mu_p, D_n, D_p, \tau_n, \tau_p$.

$$\boxed{G_{n,p} = G_{\mathrm{rad}} + G_{\mathrm{device}}}$$

Generation may include radiation-induced carriers, optical excitation, or device injection depending on the case.

Module 5: FEM weak form and carrier matrices

The drift–diffusion PDE gives a mass matrix, diffusion matrix, drift/advection matrix, recombination matrix, and source vector.

Electron weak form, schematic

$$\frac{\partial n}{\partial t} = \nabla \cdot \left(D_n \nabla n + \mu_n n \boldsymbol{E}\right) + G_n - R_n$$

$$\begin{aligned}\int_\Omega w \partial_t n \, \mathrm{d}\Omega + \int_\Omega D_n \nabla w \cdot \nabla n \, \mathrm{d}\Omega \\ - \int_\Omega \mu_n n \boldsymbol{E} \cdot \nabla w \, \mathrm{d}\Omega = \int_\Omega w (G_n - R_n) \, \mathrm{d}\Omega.\end{aligned}$$

Boundary terms encode contact injection, recombination surfaces, or insulating zero-flux edges.

Discrete form

$$\boxed{\mathbf{M}_n \dot{\boldsymbol{n}} + (\mathbf{K}_n + \mathbf{A}_n + \mathbf{R}_n)\boldsymbol{n} = \boldsymbol{g}_n}$$

$$\begin{aligned} M^e_{ab} &= \int_{\Omega_e} N_a N_b \, \mathrm{d}\Omega, \\ K^e_{ab} &= \int_{\Omega_e} D_n \nabla N_a \cdot \nabla N_b \, \mathrm{d}\Omega, \\ A^e_{ab} &= -\int_{\Omega_e} \mu_n N_b \boldsymbol{E} \cdot \nabla N_a \, \mathrm{d}\Omega. \end{aligned}$$

A parallel system is assembled for holes with the proper drift sign.

Module 5: Time stepping, currents, and verification

The transport solve advances carrier density and then postprocesses current density and degradation metrics.

Backward-Euler electron update

$$\boxed{\left[\mathbf{M}_n + \Delta t(\mathbf{K}_n + \mathbf{A}_n + \mathbf{R}_n)\right] \boldsymbol{n}^{n+1} = \mathbf{M}_n \boldsymbol{n}^n + \Delta t \boldsymbol{g}_n^{n+1}}$$

Analogous update for $\boldsymbol{p}^{n+1}$.

Current postprocessing

$$\begin{aligned} \boxed{\boldsymbol{J}_n} &\boxed{= q\mu_n n\boldsymbol{E} + qD_n \nabla n}, \\ \boxed{\boldsymbol{J}_p} &\boxed{= q\mu_p p\boldsymbol{E} - qD_p \nabla p.} \end{aligned}$$

Verification logic

- No field, no source, no recombination: uniform n, p remain constant.
- Pure recombination: recover exponential lifetime decay.
- Pure diffusion: carrier inventory conserved under zero-flux boundaries.
- Uniform field: drift/current sign matches electron and hole charge conventions.

Module 5 turns fields and material damage into measurable electrical outputs: current, carrier loss, leakage trends, and charge-collection degradation.

Module 6: Coupled multiphysics state vector

Purpose: integrate electrostatics, defect evolution, thermal transport, and carrier drift–diffusion on a shared 2D FEM mesh.

Coupled unknown fields

$$\boxed{\boldsymbol{u} = [\boldsymbol{C}_1, \ldots, \boldsymbol{C}_M, \boldsymbol{\Phi}, \boldsymbol{T}, \boldsymbol{n}, \boldsymbol{p}]^T}$$

- $\boldsymbol{C}_i$: defect concentrations.
- $\boldsymbol{\Phi}$: electrostatic potential.
- $\boldsymbol{T}$: temperature.
- $\boldsymbol{n}, \boldsymbol{p}$: carrier densities.

Shared-mesh philosophy

$\boxed{\text{same nodes/elements} \Rightarrow \text{direct field transfer}}$

When all modules use the same triangular mesh, couplings such as

$$C_i \to \rho_{\text{def}}, \quad \phi \to \boldsymbol{E}, \quad T \to D_i, k_i, \mu_n, \mu_p$$

are evaluated without interpolation errors between incompatible grids.

Module 6 is not a new physics law; it is the numerical architecture that makes Modules 2–5 talk to each other consistently.

Module 6: Coupled block residual structure

The fully coupled system can be written as interacting residual blocks.

$$\boxed{\boldsymbol{R}(\boldsymbol{u}) = \begin{bmatrix} \boldsymbol{R}_{C_1} \\ \vdots \\ \boldsymbol{R}_{C_M} \\ \boldsymbol{R}_{\phi} \\ \boldsymbol{R}_{T} \\ \boldsymbol{R}_{n} \\ \boldsymbol{R}_{p} \end{bmatrix} = \mathbf{0}}$$

Representative blocks

$$\begin{aligned} \boldsymbol{R}_{\phi} &= \mathbf{K}_{\phi}\boldsymbol{\Phi} - \boldsymbol{b}_{\rho}(\boldsymbol{C}_i, \boldsymbol{n}, \boldsymbol{p}), \\ \boldsymbol{R}_{C_i} &= \mathbf{M}_C \dot{\boldsymbol{C}}_i + \mathbf{K}_{C_i}(T, \boldsymbol{E})\boldsymbol{C}_i - \boldsymbol{g}_{C_i}, \\ \boldsymbol{R}_{T} &= \mathbf{M}_T \dot{\boldsymbol{T}} + \mathbf{K}_T \boldsymbol{T} - \boldsymbol{f}_T. \end{aligned}$$

Carrier blocks

$$\begin{aligned} \boldsymbol{R}_{n} &= \mathbf{M}_n \dot{\boldsymbol{n}} + (\mathbf{K}_n + \mathbf{A}_n(\boldsymbol{E}) + \mathbf{R}_n(C_i, T))\,\boldsymbol{n} - \boldsymbol{g}_n, \\ \boldsymbol{R}_{p} &= \mathbf{M}_p \dot{\boldsymbol{p}} + (\mathbf{K}_p + \mathbf{A}_p(\boldsymbol{E}) + \mathbf{R}_p(C_i, T))\,\boldsymbol{p} - \boldsymbol{g}_p. \end{aligned}$$

The residual view supports either a monolithic nonlinear solve or a staggered operator-splitting solve.

Module 6: Staggered execution order in MATLAB

The computer must choose an order even though the physical system is coupled.

One time step, reduced staggered loop

1. $\rho^n = q(p^n - n^n + N_D^+ - N_A^-) + q \sum_i z_i C_i^n,$
2. $\mathbf{K}_\phi \boldsymbol{\Phi}^n = \boldsymbol{b}_\rho^n, \qquad \boldsymbol{E}^n = -\nabla \phi^n,$
3. $\boldsymbol{C}_i^n \to \boldsymbol{C}_i^{n+1}$ using $T^n, \boldsymbol{E}^n,$
4. $\boldsymbol{T}^n \to \boldsymbol{T}^{n+1}$ using heat sources,
5. $\boldsymbol{n}^n, \boldsymbol{p}^n \to \boldsymbol{n}^{n+1}, \boldsymbol{p}^{n+1}$ using $\boldsymbol{E}^n, C_i^{n+1}, T^{n+1},$
6. recompute $\rho^{n+1}, \phi^{n+1}, \boldsymbol{E}^{n+1}.$

Optional inner coupling iteration

$$\boxed{\boldsymbol{u}^{n+1,k} \to \boldsymbol{u}^{n+1,k+1}}$$

Repeat the module sequence until

$$\boxed{\frac{\|\boldsymbol{u}^{k+1} - \boldsymbol{u}^k\|}{\|\boldsymbol{u}^{k+1}\|} < \epsilon_{\text{couple}}}$$

This improves consistency when field-defect-carrier feedback is strong.

Key point: defects reshape the field, and the updated field may feed back into charged-defect drift or kinetic coefficients in the next substep or iteration.

Module 6: Transfer policy, diagnostics, and summary

A coupled solver is only credible if field transfers and conservation diagnostics are explicit.

Transfer and projection rules

- Same mesh: pass nodal fields directly.
- Different mesh: project by interpolation or conservative L^2 projection.
- Element values: evaluate gradients and fluxes by element shape-function derivatives.
- Boundary data: map contact potentials, heat fluxes, and carrier fluxes consistently.

Diagnostics

$$
\begin{aligned}
Q_{\text{tot}} &= \int_\Omega \rho \, \mathrm{d}\Omega, \\
N_{C_i} &= \int_\Omega C_i \, \mathrm{d}\Omega, \\
U_T &= \int_\Omega C_T T \, \mathrm{d}\Omega.
\end{aligned}
$$

Track charge consistency, defect inventory, thermal energy, and carrier balance after each coupled step.

Final summary: Module 2 solves fields from charge; Module 3 evolves defect populations; Module 4 evolves temperature and nonlocal thermalization; Module 5 evolves carriers and current; Module 6 couples them on a shared FEM infrastructure.